# Machine Learning 101 for Financial Trading

FSE-DAT-3019-1.1 | Data Analytics and Computational Modelling

**Sector: Financial Services | Proficiency Level 3**

# LESSON PLAN

**Tertiary Infotech Academy Pte Ltd**

UEN 201200696W

**Version 2.0 | 18 October 2026**

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# Version History

| Version No | Date Effected | Change Update | Approved by |
|---|---|---|---|
| 2.0 | 18 October 2026 | Major revision, focused entirely on quant and algorithmic stock trading. Courseware rebuilt on the single-source IBF pipeline and re-platformed to Alpaca (API-first broker, paper trading, 24/5 US stock trading) and TradingView (charting, Pine Script v6, Strategy Tester, Paper Trading) with Python. Four topics: quant success stories and the evolution of quant from Thorp and Renaissance through statistical arbitrage to high-frequency trading, the paradigm shift from machine learning to LLMs and its impact on quant, and AI agents versus agentic AI in trading; the classification of trading algorithms, indicators, strategies, a TradingView versus Alpaca comparison, order placement and metric definitions; machine learning models, statistical modelling, A/B testing and pairs trading as a hedging algorithm; and an end-to-end, automated algorithmic trading pipeline. 'How the algorithm works' slides explain each algorithm on real market data. Twenty hands-on labs build one pipeline stage by stage, each with an instruction PDF containing step-by-step AI prompts, and include live paper trading of a strategy. Assessment: Written Assessment (short-answer questions, K1-K9) and Practical Performance (four tasks, A1-A10) with a standalone SMART marking rubric. Facilitator Guide added. Fee restated in SGD exclusive of GST, optimum class size stated, provider name updated to Tertiary Infotech Academy Pte Ltd. | Dr Alfred Ang |
| 1.1 | 22 December 2022 | Multiple choice questions removed; practical test retained as the sole assessment instrument. | Dr Alfred Ang |
| 1.0 | 10 June 2022 | Initial release for pilot launch, aligned to Data Analytics and Computational Modelling (Proficiency Level 3). | Dr Alfred Ang |
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*This document is at Version 2.0, effective 18 October 2026.*

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## 9. Trainer Notes

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# 1. Course Information

| Item | Details |
|---|---|
| Course Title | Machine Learning 101 for Financial Trading |
| TSC Code and Title | FSE-DAT-3019-1.1 Data Analytics and Computational Modelling |
| Sector | Financial Services |
| TSC Category | Data Analytics and Information Technology Management |
| Proficiency Level | 3 |
| Course Duration | 16 hours over 2 training days (14 hours classroom training + 2 hours assessment) |
| Daily Schedule | 0900 - 1830 (8 instructional hours per day, excluding two 15-minute tea breaks and a 1-hour lunch break) |
| Mode of Delivery | Instructor-led classroom training with hands-on practical laboratory exercises on the Alpaca paper-trading platform, TradingView and Python |
| Software Platforms | Alpaca paper-trading account (Trading API and Market Data API, alpaca-py SDK); TradingView (charts, Pine Script v6, Strategy Tester, alerts); Python 3 with pandas, NumPy, scikit-learn, XGBoost, SciPy and matplotlib |
| Assessment Methods | Written Assessment - Short Answer Questions (45 minutes) \| Practical Performance (75 minutes) |
| Training Venue | 12 Woodlands Square, #07-85/86/87, Woods Square Tower 1, Singapore 737715 |
| Class Size | Optimum class size of 20 participants (maximum 20) |
| Trainer to Learner Ratio | 1 : 20 |
| Course Fee | SGD 800 per participant (excluding GST) |

# 2. Target Audience and Entry Requirements

## Target Audience

- Traders, dealers, investment and research analysts in banks, brokerages and asset managers who want to test and automate trading ideas with data and machine learning.
- Quantitative, risk and data analysts in financial institutions who support trading, treasury or investment functions.
- Finance executives and technology professionals moving into algorithmic trading, trading technology or AI roles in financial services.
- Individual investors working in financial services who want a disciplined, evidence-based approach to systematic trading.

## Entry Requirements

### Knowledge and Skills

- Able to read, write, speak and listen in English at a proficiency level equivalent to the Employability Skills Workplace Literacy Level 5.
- Able to manipulate numbers at a proficiency level equivalent to the Employability Skills Workplace Numeracy Level 5.
- Basic computer literacy; basic familiarity with Python or spreadsheets is an advantage but not required - all code is supplied and explained.

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- Basic understanding of financial markets (stocks, ETFs, buying and selling).

**Experience**

- Preferably 1 year of working experience in financial services or a related role involving data, investment or trading.

**Equipment**

- A laptop with Python 3.10 or later, a modern web browser and internet access (set-up guidance is provided in Lab 1).
- A free Alpaca paper-trading account and a free TradingView account, created in class; no real money is used at any point.

## 3. Learning Outcomes

On completion of this course, participants will be able to:

| LO | Learning Outcome | TSC Reference |
|---|---|---|
| LO1 | Develop hypotheses for implementing data analytics and computational modelling activities in algorithmic trading, and identify possible big data applications across the forms of data within financial services. | A1, A6, K1, K9 |
| LO2 | Identify appropriate statistical algorithms and data models to test trading hypotheses, and use appropriate analytics platforms, analytical tools and statistical methods given specific analytics and reporting requirements. | A2, A3, K3, K5, K7 |
| LO3 | Conduct statistical modelling of market data with machine learning models to derive patterns and solutions, and perform A/B testing of trading strategies based on different hypotheses. | A4, A5, K2, K4, K6 |
| LO4 | Perform coding and configuration of an automated trading program based on selected models or algorithms, conduct tests on its actions and outcomes, diagnose unintended outcomes, and draw insights from the results to support decisions and report accordingly. | A7, A8, A9, A10, K7, K8 |

## 4. TSC Alignment

This course is fully aligned to FSE-DAT-3019-1.1 Data Analytics and Computational Modelling at Proficiency Level 3: "Identify and utilise appropriate statistical algorithms and data models to perform data analyses, test hypotheses and derive inferences, patterns or solutions from data". All 10 ability statements and all 9 knowledge statements of the TSC are delivered and assessed.

### Abilities

| Ref | Ability Statement | Delivered In | Assessed By |
|---|---|---|---|
| A1 | Develop hypotheses for implementing data analytics and computational modelling activities | Topic 1 | PP Task 1 |
| A2 | Identify appropriate statistical algorithms and data models to test hypotheses and/or theories | Topics 2, 3 | PP Task 2 |
| A3 | Use appropriate analytics platforms, analytical tools, statistical methods and analytical approaches given specific analytics and reporting requirements | Topics 1, 2 | PP Task 2 |

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| Ref | Ability Statement | Delivered In | Assessed By |
|---|---|---|---|
| A4 | Conduct statistical modelling of data to derive patterns and solutions | Topic 3 | PP Task 3 |
| A5 | Perform A/B testing based on different hypotheses | Topic 3 | PP Task 3 |
| A6 | Identify possible big data applications | Topic 1 | PP Task 1 |
| A7 | Perform coding and configuration of software agents or programs based on selected models or algorithms | Topic 4 | PP Task 4 |
| A8 | Conduct tests on the actions taken and outcomes to assess effectiveness of the models | Topic 4 | PP Task 4 |
| A9 | Diagnose unintended outcomes produced by statistical models | Topic 4 | PP Task 4 |
| A10 | Draw relevant trends and insights from data analysis to support decisions and report accordingly | Topic 4 | PP Task 4 |

### Knowledge

| Ref | Knowledge Statement | Delivered In | Assessed By |
|---|---|---|---|
| K1 | Data analytics and modelling business use cases | Topics 1, 4 | WA Q1 |
| K2 | Types of statistical analyses, data models, algorithms and advanced computational methods | Topic 3 | WA Q2 |
| K3 | Range and application of various statistical methods and algorithms | Topic 2 | WA Q3 |
| K4 | Range and application of various types of data models | Topic 3 | WA Q4 |
| K5 | Usage of analytics platforms and tools | Topics 1, 2, 4 | WA Q5 |
| K6 | Statistical modelling techniques | Topic 3 | WA Q6 |
| K7 | Coding languages for programming of algorithms and signals | Topics 2, 4 | WA Q7 |
| K8 | Potential reasons for unintended outcomes | Topic 4 | WA Q8 |
| K9 | Forms of data within financial services | Topic 1 | WA Q9 |

## 5. Course Structure and Duration by Topic

| Topic | Title | Day | Labs | TSC Ref | Duration | Slides |
|---|---|---|---|---|---|---|
| Topic 1 | Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents | Day 1 | Labs 1-4 | A1, A6, K1, K9 | 3 hours | 28 - 66 |
| Topic 2 | Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca | Day 1 | Labs 5-9 | A2, A3, K3, K5, K7 | 4.25 hours | 67 - 104 |
| Topic 3 | Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals | Day 2 | Labs 10-14 | A4, A5, K2, K4, K6 | 2.5 hours | 105 - 133 |
| Topic 4 | Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis | Day 2 | Labs 15-20 | A7, A8, A9, A10, K7, K8 | 3.25 hours | 134 - 155 |
| Admin | Course overview, ground rules, daily recaps | 1, 2 | - | - | 1 hour | 1 - 27 |
| Assessment | Written Assessment and Practical Performance | 2 | - | K1-K9, A1-A10 | 2 hours | - |
| TOTAL | | | | | 16 hours | |

*Topic hours include the trainer-led content and the hands-on labs for that topic. Training 14 hours + assessment 2 hours = 16 hours, excluding tea and lunch breaks.*

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## Hands-on labs

| Lab | Title | Topic | TSC Ref | Duration | Slide |
|---|---|---|---|---|---|
| 1 | Pipeline Skeleton, Configuration and the Alpaca Paper Account | 1 | K5 | 20 min | 58 |
| 2 | Data-Validation Stage: Cross-Check Pipeline Bars against TradingView | 1 | A3, K5, K9 | 20 min | 59 |
| 3 | Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality | 1 | A6, K9 | 20 min | 61 |
| 4 | Hypothesis Stage: from a Written Card to an Automated Statistical Test | 1 | A1, A6, K1 | 15 min | 64 |
| 5 | Indicator Stage: Pine Script v6 Indicators and their Python Twins | 2 | A3, K3, K7 | 25 min | 93 |
| 6 | Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation | 2 | A2, A3, K3, K7 | 30 min | 95 |
| 7 | Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals | 2 | A2, K3, K7 | 25 min | 98 |
| 8 | Execution Stage: an Order Manager that Trades the Paper Account in Any Session | 2 | A3, K5 | 30 min | 100 |
| 9 | Evaluation Stage: Performance and Risk Metrics with a Benchmark Table | 2 | A3, K3 | 25 min | 101 |
| 10 | Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay | 3 | A2, A4, K2 | 20 min | 123 |
| 11 | ML-Signal Stage: a Walk-Forward Classifier from Probability to Position | 3 | A4, K4, K6 | 20 min | 125 |
| 12 | Regime-Filter Stage: PCA and K-Means as a Gate on the Signals | 3 | A4, K4, K6 | 15 min | 127 |
| 13 | A/B-Test Stage: a Statistical Decision before Capital Moves | 3 | A5, K2 | 15 min | 129 |
| 14 | Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve | 3 | A4, K6 | 20 min | 131 |
| 15 | Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even | 4 | A7, A8, K7 | 25 min | 144 |
| 16 | Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading | 4 | A7, A8, K7 | 25 min | 146 |
| 17 | Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution | 4 | A7, K5, K7 | 20 min | 148 |
| 18 | Diagnostics Stage: Automated Bias Checks before Go-Live | 4 | A9, K8 | 20 min | 149 |
| 19 | Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report | 4 | A8, A10 | 25 min | 151 |
| 20 | Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval | 4 | A7, A10, K1 | 20 min | 153 |

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## 6. Daily Training Schedule

Each training day comprises 8 instructional hours from 0900 to 1830, excluding two 15-minute tea breaks and a 1-hour lunch break. The 2-hour final assessment (45-minute Written Assessment and 75-minute Practical Performance) is conducted at the end of Day 2.

### Day 1

| Time | Topic / Activity | Learning Outcome | Delivery Method & Resources | TSC Ref |
|---|---|---|---|---|
| 09:00 - 09:30 | Course Overview, Trainer Introduction and Ground Rules<br>Course objectives, TSC alignment, assessment overview, ground rules and an icebreaker on how participants use data in trading or investment today. | - | Trainer presentation; Slides; Learner Guide | - |
| 09:30 - 10:45 | T1.1-1.3 History of AI, AI Agents versus Agentic AI, and the History of Algorithmic Trading<br>Milestones from the Turing Test to agentic AI; AI agents versus agentic AI and multi-agent trading systems; the six paradigm shifts of algorithmic trading. | LO1 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | K1, K9 |
| 10:45 - 11:00 | Tea Break | | | |
| 11:00 - 11:30 | T1.4-1.5 Forms of Financial Data, Big Data Applications and Trading Hypotheses<br>Market, reference, fundamental and alternative data; the four Vs; big data applications; writing a testable hypothesis. | LO1 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A1, A6, K1, K9 |
| 11:30 - 11:50 | Lab 1: Pipeline Skeleton, Configuration and the Alpaca Paper Account<br>Create the pipeline package and its config.yaml (universe, strategy parameters, risk limits, costs, secret names), build a validating configuration loader and a market-clock module, and run `python -m pipeline check` against the Alpaca paper account or the offline simulator. | LO1 | Hands-on lab; Learner Guide; lab folder Lab01-alpaca-setup | K5 |
| 11:50 - 12:10 | Lab 2: Data-Validation Stage: Cross-Check Pipeline Bars against TradingView<br>Read daily bars for SGX:D05 and NASDAQ:AAPL from the TradingView Data Window, record them, and build a validation stage that compares TradingView with the pipeline's raw and dividend-adjusted bars and writes a validation report. | LO1 | Hands-on lab; Learner Guide; lab folder Lab02-tradingview-charts | A3, K5, K9 |
| 12:10 - 12:30 | Lab 3: Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality<br>Build `python -m pipeline ingest` to fetch and cache daily, weekly, monthly and hourly bars for the universe, run automated data-quality checks, ingest headlines as unstructured data, and describe the forms of data and the 3 Vs of big data. | LO1 | Hands-on lab; Learner Guide; lab folder Lab03-alpaca-market-data | A6, K9 |
| 12:30 - 13:30 | Lunch Break | | | |
| 13:30 - 13:45 | Lab 4: Hypothesis Stage: from a Written Card to an Automated Statistical Test<br>Write a falsifiable hypothesis card in YAML (H1, H0, signal known at the close, outcome, test, decision rule, big-data use case) and build `python -m pipeline hypothesis` to test it with a Welch t-test, a bootstrap confidence interval and a robustness grid. | LO1 | Hands-on lab; Learner Guide; lab folder Lab04-trading-hypothesis | A1, A6, K1 |

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| Time | Topic / Activity | Learning Outcome | Delivery Method & Resources | TSC Ref |
|---|---|---|---|---|
| 13:45 - 14:30 | T2.1-2.2 Technical Indicators and Standard Algorithmic Trading Strategies<br>SMA, EMA, RSI, MACD, Bollinger Bands, ATR, VWAP, OBV; trend following, mean reversion, momentum, breakout, pairs trading and execution algorithms. | LO2 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A2, K3 |
| 14:30 - 14:55 | Lab 5: Indicator Stage: Pine Script v6 Indicators and their Python Twins<br>Add the desk's Pine Script v6 indicators to a TradingView chart, build pipeline/indicators.py with twins of ta.ema, ta.sma, ta.rsi, ta.bb, ta.atr, ta.macd, ta.crossover and ta.crossunder, and cross-check the values against the TradingView Data Window. | LO2 | Hands-on lab; Learner Guide; lab folder Lab05-pine-indicators | A3, K3, K7 |
| 14:55 - 15:25 | Lab 6: Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation<br>Configure a Pine Script v6 strategy with costs, read the Strategy Tester, build `python -m pipeline reconcile` to check the Pine settings and reproduce the results, and place and manage paper trades from the chart with TradingView Paper Trading (and optionally an Alpaca paper account). | LO2 | Hands-on lab; Learner Guide; lab folder Lab06-pine-strategy-tester | A2, A3, K3, K7 |
| 15:25 - 15:40 | Tea Break | | | |
| 15:40 - 16:05 | Lab 7: Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals<br>Build pipeline/signals.py with trend, mean-reversion, momentum and breakout signals that return target positions with next-bar alignment, a registry, an automated look-ahead test and a universe target file for the execution stage. | LO2 | Hands-on lab; Learner Guide; lab folder Lab07-python-indicators | A2, K3, K7 |
| 16:05 - 16:50 | T2.3-2.4 Analytics Platforms, Coding Languages and Order Placement<br>TradingView, Alpaca Trading and Market Data APIs, Python libraries; Pine Script and Python; order types, time in force, bracket orders and the order lifecycle. | LO2 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A3, K5, K7 |
| 16:50 - 17:20 | Lab 8: Execution Stage: an Order Manager that Trades the Paper Account in Any Session<br>Build pipeline/execution.py: a session-aware, idempotent order manager that sizes orders from target weights, sends market, extended-hours limit, GTC-queued or bracket PAPER orders, polls their lifecycle, cancels protective legs before exits, and falls back safely when an order is rejected. | LO2 | Hands-on lab; Learner Guide; lab folder Lab08-paper-orders | A3, K5 |
| 17:20 - 17:50 | T2.5 Performance and Risk Metric Definitions<br>CAGR, volatility, Sharpe, Sortino, Calmar, maximum drawdown, win rate, profit factor, expectancy, exposure, turnover, alpha and beta. | LO2 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A3, K3 |
| 17:50 - 18:15 | Lab 9: Evaluation Stage: Performance and Risk Metrics with a Benchmark Table<br>Build `python -m pipeline evaluate` to compute total return, CAGR, volatility, Sharpe, Sortino, maximum drawdown, Calmar, win rate, profit factor, exposure, turnover, alpha and beta for every signal against a buy-and-hold benchmark, and interpret the ranking. | LO2 | Hands-on lab; Learner Guide; lab folder Lab09-performance-metrics | A3, K3 |
| 18:15 - 18:30 | Day 1 Recap and Q&A<br>Consolidation of Topics 1 and 2; preview of Day 2. | - | Trainer presentation; Slides; Learner Guide | - |

**Total instructional time, Day 1: 8 hours (480 minutes), excluding the tea breaks and the 1-hour lunch break.**

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## Day 2

| Time | Topic / Activity | Learning Outcome | Delivery Method & Resources | TSC Ref |
|---|---|---|---|---|
| 09:00 - 09:15 | Day 1 Recap and Day 2 Overview<br>Review of Topics 1 and 2; orientation to Topics 3 and 4 and to the assessment. | - | Trainer presentation; Slides; Learner Guide | - |
| 09:15 - 09:40 | T3.1-3.2 Statistical Analyses, Algorithms and Supervised Learning Models<br>Descriptive to prescriptive analysis; supervised, unsupervised and reinforcement learning; strategy search; regression, trees, random forest, XGBoost, neural networks. | LO3 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A4, K2, K4 |
| 09:40 - 10:00 | Lab 10: Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay<br>Build `python -m pipeline search` to grid-search three signal families, rank parameter sets in sample, measure out-of-sample decay, prefer a robust parameter region, and decide whether anything deserves promotion into config.yaml. | LO3 | Hands-on lab; Learner Guide; lab folder Lab10-strategy-grid-search | A2, A4, K2 |
| 10:00 - 10:20 | Lab 11: ML-Signal Stage: a Walk-Forward Classifier from Probability to Position<br>Build pipeline/ml.py: engineer causal features, train logistic regression, random forest and XGBoost with walk-forward TimeSeriesSplit validation, turn P(up) into a target position and backtest it after costs. | LO3 | Hands-on lab; Learner Guide; lab folder Lab11-ml-signal-classifier | A4, K4, K6 |
| 10:20 - 10:35 | Tea Break | | | |
| 10:35 - 10:55 | T3.3-3.4 Unsupervised Learning and Statistical Modelling Techniques<br>K-means, hierarchical clustering and PCA for regimes; returns and stationarity; time-ordered validation; feature engineering; regularisation. | LO3 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A4, K4, K6 |
| 10:55 - 11:10 | Lab 12: Regime-Filter Stage: PCA and K-Means as a Gate on the Signals<br>Build pipeline/regime.py: PCA on the universe's returns, K-means regimes on backward-looking SPY features, walk-forward regime labels, and a gate that forces signals flat in the stress regime, evaluated after costs. | LO3 | Hands-on lab; Learner Guide; lab folder Lab12-market-regimes | A4, K4, K6 |
| 11:10 - 11:25 | T3.5-3.6 A/B Testing, and Pairs Trading as a Hedging Algorithm<br>Hypotheses, Welch's t-test, bootstrap Sharpe difference, multiple testing; cointegration, hedge ratio, spread z-score and market neutrality. | LO3 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A4, A5, K2, K6 |
| 11:25 - 11:40 | Lab 13: A/B-Test Stage: a Statistical Decision before Capital Moves<br>Build `python -m pipeline abtest` to compare any two library signals with a Welch t-test, a paired t-test and a block-bootstrap confidence interval for the Sharpe difference, and to record an accept or reject decision with a multiple-testing caution. | LO3 | Hands-on lab; Learner Guide; lab folder Lab13-ab-testing | A5, K2 |
| 11:40 - 12:00 | Lab 14: Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve<br>Build pipeline/pairs.py: screen pairs with the Engle-Granger cointegration test, estimate the hedge ratio, trade the spread z-score out of sample with next-bar execution and costs on both legs, confirm market neutrality and output leg targets. | LO3 | Hands-on lab; Learner Guide; lab folder Lab14-pairs-trading | A4, K6 |
| 12:00 - 12:30 | T4.1-4.2 Pipeline Architecture and Backtesting with Trading Costs<br>The eight-stage pipeline; configuration and pre-trade risk controls; next-bar execution; commission, spread, slippage and regulatory fees. | LO4 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A7, A8, K7 |
| 12:30 - 13:30 | Lunch Break | | | |

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| Time | Topic / Activity | Learning Outcome | Delivery Method & Resources | TSC Ref |
|---|---|---|---|---|
| 13:30 - 13:55 | Lab 15: Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even<br>Build pipeline/backtest.py around the common backtester: run any library signal under every cost scenario (and a custom one), break the cost into commission, spread, slippage and regulatory fees, find each strategy's break-even cost and test next-open versus next-close execution. | LO4 | Hands-on lab; Learner Guide; lab folder Lab15-trading-costs | A7, A8, K7 |
| 13:55 - 14:05 | T4.3 Forward Testing with Automated Paper Trading<br>The bot loop, market clock, risk checks, order status and logging. | LO4 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A7, A8 |
| 14:05 - 14:30 | Lab 16: Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading<br>Build pipeline/risk.py and pipeline/runner.py: `python -m pipeline run --once \| --loop` with the market clock, risk checks, kill switch and trade log; run a 2025 forward test; run the strategy live on the paper account for a few cycles; and schedule it with cron, launchd or Windows Task Scheduler. | LO4 | Hands-on lab; Learner Guide; lab folder Lab16-paper-trading-bot | A7, A8, K7 |
| 14:30 - 14:50 | Lab 17: Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution<br>Build pipeline/webhook.py: a webhook receiver that turns TradingView alert JSON into a pipeline decision (validation -> signal -> risk -> execution), test it with a local alert simulator and a self-test, and configure a real TradingView strategy alert. | LO4 | Hands-on lab; Learner Guide; lab folder Lab17-tradingview-webhook | A7, K5, K7 |
| 14:50 - 15:00 | T4.4 Potential Reasons for Unintended Outcomes<br>Overfitting, look-ahead and survivorship bias, cost drag, regime change, data and software errors; the diagnosis method. | LO4 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A9, K8 |
| 15:00 - 15:20 | Lab 18: Diagnostics Stage: Automated Bias Checks before Go-Live<br>Build `python -m pipeline diagnose` to run automated look-ahead (features and execution), cost-drag, overfitting and survivorship checks on any strategy and on the junior quant's exhibit, and write a PASS / WARN / FAIL report with a go-live verdict. | LO4 | Hands-on lab; Learner Guide; lab folder Lab18-diagnose-bias | A9, K8 |
| 15:20 - 15:35 | Tea Break | | | |
| 15:35 - 15:45 | T4.5 Performance Analysis, Insights and Reporting<br>Tearsheets, backtest versus forward test, insights and the recommendation. | LO4 | Trainer presentation, discussion and worked examples; Slides; Learner Guide | A10 |
| 15:45 - 16:10 | Lab 19: Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report<br>Build `python -m pipeline report` to compare the research backtest with the runner's forward-test logs, run monitoring checks with thresholds, and generate a tearsheet and a markdown report with data-driven insights and a recommendation. | LO4 | Hands-on lab; Learner Guide; lab folder Lab19-performance-report | A8, A10 |
| 16:10 - 16:30 | Lab 20: Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval<br>Build pipeline/agents.py (analyst, researcher, trader and risk agents; rule-based by default, Claude optional) and pipeline/e2e.py (`python -m pipeline e2e`) that runs every stage in order with gates, then evaluate the pipeline and the role of human oversight. | LO4 | Hands-on lab; Learner Guide; lab folder Lab20-agentic-capstone | A7, A10, K1 |
| 16:30 - 17:15 | Written Assessment (Short Answer Questions)<br>9 open-ended short-answer questions covering K1 to K9. | All LOs | Individual assessment; Assessment papers | K1 - K9 |

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| Time | Topic / Activity | Learning Outcome | Delivery Method & Resources | TSC Ref |
|---|---|---|---|---|
| 17:15 - 18:30 | Practical Performance Assessment<br>4 practical tasks on Python, the Alpaca paper-trading API and the supplied dataset, covering A1 to A10. | All LOs | Individual assessment; Assessment papers | A1 - A10 |

**Total instructional time, Day 2: 8 hours (480 minutes) = 6 hours training + 2 hours assessment, excluding the tea breaks and the 1-hour lunch break.**

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## 7. Assessment Summary

Participants must be assessed as Competent (C) in BOTH assessment components to be awarded competency for this module. Each component is marked out of 100 against the marking points in the Assessment Rubric; Competent requires at least 80 marks in each component, at least 60% of the marks for every knowledge and ability statement, and every critical marking point awarded.

| Method | Duration | Format | TSC Ref | Coverage |
|---|---|---|---|---|
| Written Assessment (Short Answer Questions) | 45 minutes | 9 open-ended short answer questions | K1 - K9 | All 9 knowledge statements |
| Practical Performance | 75 minutes | 4 practical tasks in Python with the Alpaca paper-trading API (or the supplied simulator) on a supplied dataset | A1 - A10 | All 10 ability statements |

Assessment format: Open book. Candidates may refer to their Learner Guide, the lab code and their own class notes. Refer to the Assessment Plan for the full assessment specification, marking guides and records.

## 8. Training Resources

### Trainer Resources

- Trainer slide deck (PowerPoint), 161 slides covering all four topics and twenty labs.
- Lesson Plan (this document) and Facilitator Guide.
- Assessment Plan, including question papers, marking guides, rubric and Assessor Key.
- Lab folder (labs) with all scripts, Pine Script files, bundled market data and reference outputs; a trainer Alpaca paper account for live demonstrations.
- Laptop with Python 3.10+, a modern web browser, projector and whiteboard.

### Learner Resources

- Learner Guide containing all topic content and step-by-step instructions for every lab.
- One laptop per participant with Python 3.10+, a modern web browser and internet access; a free Alpaca paper-trading account and a free TradingView account (created in Lab 1 and Lab 2).
- The lab folder, including requirements.txt, the common library and bundled daily market data so every lab also runs offline.

## 9. Trainer Notes

- Paper trading only. Never demonstrate with a live-money account, and remind participants that nothing in the course is investment advice.
- If the venue network blocks the Alpaca API, or a participant cannot open an account, every lab runs offline on the bundled data and the built-in paper-broker simulator. Show both paths early (Lab 1) so nobody is blocked.
- Webhook alerts on TradingView need a paid plan. In Lab 17 demonstrate with the trainer's account if available, and let participants use send_test_alert.py, which posts the identical JSON a TradingView alert sends.

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- Lab 18 contains deliberately seeded faults. Do not reveal them; let participants find the look-ahead first, because it produces no error message - only a result that is too good.
- Keep the finance context foremost. When a participant reports a great backtest, ask the three questions: is it net of costs, is it out of sample, and what would the investment committee need to see before risking capital?
- In Topic 1, link every milestone of the AI and algorithmic-trading histories to a control used later in the course (circuit breakers to the kill switch, Knight Capital to deployment testing, MiFID II to algorithm records).

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